Survey of Data and Storage Security in Cloud Computing

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Abstract- **Data Security and consumer data privacy are the key challenges in cloud computing era. The appropriateness and privacy of data stored in cloud may be compromised because of limited security for data owners. This paper presents an extensive survey on privacy preservation, data and storage security challenging issues in cloud computing. The Security of cloud data is further analyzed in terms of data integrity, access control and attribute based encryption. The survey analyzes each category of work in detail. A comparison table is also presented along with the strength and weakness of each approach.**

Keyword- **Cloud Computing, Data, Storage, Security, Servers.**

I. INTRODUCTION

Cloud computing is an open standard model which can enable ubiquitous computing and provide network access to a shared group of configurable computing resources. It provides distributed computing environment consisting hardware, software and services [1]. Apart from this, it provides storage space and support execution of various services and data processing [2]. Cloud security handles the weakness and susceptibility of cloud computing [3]. Cloud security can be categorized as: Cloud Data Security and Storage Security. Data security ensures the privacy and confidentiality of shared data while the storage security ensures the correctness of the uploaded data stored in untrustworthy cloud servers. But cloud computing has many challenges regarding both data and storage security [4].

A. Data Security

While sharing the data in cloud space, the cloud service provider (CSP) can completely acquire access to all user data. So, data sharing bring challenges in terms of security in cloud computing [26]. The main privacy and security requirements of data sharing in cloud are: Data Confidentiality and User revocation [29].

B. Storage Security

Cloud Data Owners (CDOs) upload their local data into the cloud. If some CSPs are dishonest, they may conceal the data loss or error from the users for their own benefit. Moreover there are chances that they might delete rarely accessed user data for saving storage space. Therefore, CDOs need to be ensured about the correctness of data stored in the cloud [5] [6]. Another challenge is verifying the integrity of stored data [7].

C. Organization of the Paper

This paper presents an extensive survey of data and storage security mechanisms in cloud by studying and analyzing the existing approaches along with their strengths and weakness. Moreover in this paper the existing survey on cloud security is presented in Section II, and Section III is presented with the survey of existing security solution on cloud computing which is followed by the comparison table for the existing security works in section IV. The paper is concluded in Section V.

II. SURVEY ON CLOUD SECURITY

Some of the existing survey works on security issues of cloud computing is discussed below.

Naresh vurukonda ets al [10] have made a study which identified the issues of cloud data storage, identity management and access control. Possible solutions were suggested for some of the issues.

Ayesha Malik et al [11] have defined a methodology for cloud providers that protect users' data and important information. In their study, they have explained different characteristics of the cloud computing, different service models etc.

Yunchuan Sun et al [12] have reviewed different security solutions for data storage security and privacy protection in cloud computing. They have

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presented a comparative research analysis of the existing techniques regarding data security

Mazhar Ali et al [13] have discussed various security issues of cloud computing.. Their survey consists of latest security solutions along with complete discussion on security issues. They also provided a brief discussion on security issues and solutions related to mobile cloud computing.

Sultan Aldossary et al [14] have discussed the issues of cloud data storage and solutions. The survey included issues of virtualization, data integrity, data availability, data confidentiality. Apart from these data security issues, they have listed out various threats on cloud computing.

III. SURVEY OF DATA AND STORAGE SECURITY

A. Classification of Security in Cloud computing

As shown in Figure 1, the data Security issue in Cloud Computing able to be classified.

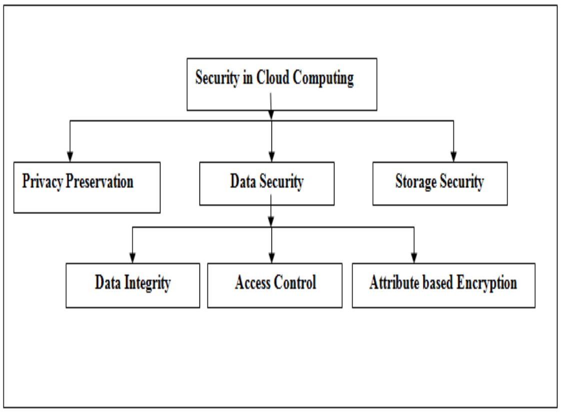

Fig.1. Security Classification in Cloud.

Furthermore data security is classified into three different categories.

- Privacy Preservation defines that Privacy of personal and important information in cloud is crucial as the cloud servers are not trusted. Confidentiality and authorization are main requirements of privacy preservation in cloud.

- Storage Security defines that Cloud storage Security is the task of providing integrity to the shared data stored at dishonest cloud servers.

- Data Security defines that the data or information security is the process of protecting the data from unauthorized users, preventing alterations and restricting the access of sensitive information.

B. Existing works on Privacy preservation

Haralambos Mouratidis et al [9] have presented a security framework for selecting a cloud provider based on relevant security requirements. It consists of a modelling language and a tool using the Open Models Initiative (OMI) Platform. The language applies the principles of security and privacy.

L. Malina et al [15] have proposed a security model for cloud based on group signatures. It provides authentication and anonymity for user privacy. It also ensures confidentiality and integrity of transmitted data.

Ulrich Greveler et al [18] have designed a cloud database architecture that prevents unauthorized access of uploaded data from the internal and external administrators. It uses XACML structure for describing the access control policy. The contents of the database are encrypted using an Encryption Proxy.

C. Existing works on Storage Security

Kan Yang et al [6] have designed a privacy-preserving auditing protocol for cloud systems. The protocol supports dynamic operations on data and batch auditing for multi-cloud environment. It uses bilinear pairing to generate an encrypted proof. The verification of the proof correctness is then executed by the data auditor. In this protocol, the computational overhead of the auditor is moved to the cloud server. However it fails to provide data confidentiality and user authorization.

Qian Wang et al [7] have a provided a verification scheme for storage security by integrating data integrity and dynamic data operations. In this scheme, an auditor verifies the integrity of storage data. The dynamic data operations include block insertion and deletion using Merkle Hash Tree (MHT) technique. They have applied the technique of bilinear aggregate signature for maintaining multiple auditing tasks. But it does not provide confidentiality and authorization.

Yan Zhu et al [8] have designed a Provable Data Possession (PDP) scheme for ensuring the integrity of cloud data storage. In PDP, multiple cloud service providers (CSPs) cooperatively maintain the client's data. The two main components of PDP are hierarchy using hash index and homomorphic verifiable response. It provides defence against leakage of data and forgery of tag attacks. But it does not provide confidentiality and authorization.

Lifei Wei et al [19] have proposed SecCloud which provides both secure storage and computation auditing features. By using verifier signature and batch verification techniques, it achieves privacy cheating discouragement. But it does not provide confidentiality and authorization.

D. Existing works on Data Security

The three important characteristics of data security are listed below.

- Data Integrity: It is commonly assured by validations using cryptographic tools such as message digests, hashing and digital signature etc.
- Access Control (AC): An access control system includes components and methods to specify access control policies for legitimate users.
- Attribute based Encryption (ABE): The uploaded data is encrypted using ABE that defines access policy on attributes related to the data. Hence only authorized users with matching attributes can decrypt and access the data.

a. Existing works on Data Integrity

Laicheng Cao et al [5] have proposed Mobile Multi Cloud Data Integrity Verification scheme (MMCDIV). This scheme support dynamic data operations in MMC and integrity verification by means of lightweight computing techniques. But it does not provide confidentiality and authorization. Moreover, it depends on single TTP server, which may be subjected to failures.

Nedhal A. Al-Saiyd et al [20] have designed a cloud computing security model along with the discussion of cloud security threats and risks. It also discuss about various security techniques which provide solution to the security threats on cloud computing. But it fails to present any standard algorithm or technique for providing data integrity.

Ali Mohammed Hameed Al-Saffar et al [21] have proposed a framework for provable data possession (PDP) in the distributed multi-cloud environment. It consists of many cloud servers to which the client's requests are sent. It ensures data integrity in multiple clouds. But how the verifier and user got their decryption keys and the process of splitting the data are not clearly explained.

Edoardo Gaetani et al [22] have applied Blockchain technology for providing data integrity. In this paper, the case study of European SUNFISH project is considered for which a blockchain-based database is designed. But the operations mainly depend on database model which may incur huge storage overhead.

b. Existing works on Access Control

Younis A. Younis et al [1] have proposed a novel Access Control model for Cloud Computing (AC3) to satisfy the requirements of access control. It consists of three hierarchical levels of security based on the level of trust. In this model, users are classified according to their roles and assigned to corresponding security domains. But this work does not provide confidentiality and integrity.

Jin Li et al [2] have designed a fine-grained access control system based on ABE. AC policies are defined based on the data attributes. The accountability of user is implemented by applying the traitor tracing method. Furthermore user revocation and user grant operations are implemented by applying a broadcast encryption technique. But it results in huge communication overhead.

Yan Zhu et al [4] have presented encryption scheme for temporal access control (TACE). In this algorithm, the access policy contains temporal attributes of users based on which the access rights are defined. TACE enforces temporal constraints. But this work does not provide confidentiality and integrity.

Guoyuan Lin et al [23] have provided a trust-based AC technique for cloud computing. It defines trust connections among users and cloud platform. It combines role-based access control (RBAC) with trust model. The trust model is formed by combining identity trust and behavior trust. But this work does not provide confidentiality and integrity.

Varsha D. Mali et al [24] have designed a trust-based cloud storage system in which trust model is integrated with RBAC technique. But this work does not provide integrity and authentication.

c. Existing works on ABE

Saravana Kumar et al [3] have proposed a new encryption technique based on ABE. It uses digital signature and asymmetric encryption algorithms with hash functions. But since the encryption technique is based on simple hash functions, it can be compromised.

Shulan Wang et al [25] have proposed a file hierarchy based ABE scheme for cloud computing. In this scheme, hierarchical files are encrypted using an integrated access structure. The ciphertext portions of attributes are shared by the files. But it does not provide data integrity. Moreover, it depends on single TA which may be subjected to failure.

Shulan Wang et al [26] have proposed an improved two-party key distribution protocol. In this protocol, any user's secret key cannot be compromised by either the key authority or CSP. Moreover, they have included weights for each attributes to enhance the expression of attribute from binary to arbitrary level. Because of this, the storage cost and encryption cost are reduced. But the access policy is not hidden and the single TA may be subjected to failure.

Tran Viet Xuan Phuong et al [27] have proposed CP-ABE scheme. In this scheme, AND-gate with wildcards is used to define the access policy. The access policy is protected using hidden ciphertext policy. However, the key escrow problem is not resolved.

Entao Luo et al [28] have proposed a hierarchical multi-authority and CB-ABE based friend discovery scheme. It uses character attribute subsets to avoid single point failure and performance overhead. But this work does not provide data integrity.

IV. COMPARISON OF EXISTING WORKS

A comparative study shown below in Table 1, reveals the proposed scheme, services, privacy, confidentiality, integrity, access control and storage from 21 existing paper works by different authors.

Authors	**Method**	**Services**	**Privacy**	**Confidentiality**	**Integrity**	**Access Control**	**Storage security**
Younis A. Younis et al	Cloud computing using Access Control Architecture	Secure access permission for multiple services	No	No	No	Yes	No
Varsha D. Mali et al	User Authentication and Access Control Technique in Cloud Computing	Cryptographic RBAC	Yes	Yes	No	Yes	No
Jin Li et al	Fine-grained Data Access Control Systems with User Accountability	ABE	No	Yes	No	Yes	No
Guoyuan Lin et al	Trust Based AC Policy	Trust Management, role-based access control	No	No	No	Yes	No
Yan Zhu et al	Temporal Access Control	Temporal access control Encryption, proxy-based re-encryption	No	Yes	No	Yes	No
Saravana Kumar et al	Enhanced ABE	ABE using digital signature and asymmetric encryption	No	Yes	Yes	Yes	No
Shulan Wang et al	An Efficient File Hierarchy ABE Scheme	ABE using Integrated access structure	No	Yes	No	Yes	No
Shulan Wang et al	Attribute-Based Data Sharing Scheme	Two-party key issuing protocol, weighted	No	Yes	No	Yes	No

		attributes					
Tran Viet Xuan Phuong et al	Hidden Ciphertext Policy	ABE using hidden access policy	No	Yes	No	Yes	No
Entao Luo et al	Hierarchical Multi-Authority and ABE Friend Discovery Scheme	ABE using multi authority, friend discovery schemes	No	Yes	No	Yes	No
Nedhal A. et al	Data Integrity In Cloud Computing Security	Data Integrity checking algorithm	No	No	Yes	No	No
Laicheng Cao et al	Data Integrity Verification Scheme	Mobile Multiple cloud Data integrity Verification	No	No	Yes	No	No
Ali Mohammed Hameed Al-Saffar et al	Identity Based technique	Data integrity and Confidentiality in the multiple cloud environment.	No	Yes	Yes	No	No
Edoardo Gaetani et al	Block chain-based Database to Ensure Data Integrity	Integrity verification using Block chain database	No	No	Yes	No	No
L. Malina et al	Privacy-preserving security solution	Non bilinear group signature scheme	Yes	Yes	Yes	No	No
Haralambos Mouratidisa et al	A framework to support selection of cloud providers	Modelling language and selection of CSP	Yes	No	No	No	No
Ulrich Greveler et al	Cloud Computing with a System that Preserves Privacy	Trust model prevents CSP from accessing outsourced data.	Yes	No	Yes	Yes	No
Lifei Wei et al	Security and privacy for storage and computation	Discouragement of Privacy cheating and auditing of secure computation	Yes	No	Yes	No	Yes
Yan Zhu et al	Cooperative Provable Data Possession for Integrity Verification	Homomorphic verifiable response and hash index hierarchy	No	No	Yes	No	Yes
Qian Wang et al	Auditing and Data Dynamics for Storage Security	verify the integrity of the dynamic data stored in the cloud	No	No	Yes	No	Yes
Kan Yang et al	Secure Dynamic	privacy-preserving	Yes	No	Yes	No	Yes

	Auditing Protocol	auditing protocol					

Table 1: Comparison of Existing Works Based on Data and Storage Security in Cloud Computing

V. CONCLUSION

In this survey work, the data security and storage security in cloud computing have been explained in detail. The different classifications in data and storage security are explained with the proposed techniques. Finally a comparison table is presented with all the existing methods tabulated with advantages and drawbacks. From the table it can be concluded that both data and storage security should be provided with less storage and computational overhead. In data security, authentication, authorization confidentiality and integrity should be ensured.

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